

Siddharth Jain

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EDUCATION

Ph.D. in Physics, University of Pittsburgh (GPA 3.8/4.0) *Aug 2023 – present*

M.Sc. in Physics, Indian Institute of Technology, Kanpur (CGPA 8.56/10) *Aug 2020 – May 2022*

B.Sc. (Hons.) in Physics, Institute of Science, BHU (CGPA 8.53/10) *Jul 2017 – Sep 2020*

RESEARCH EXPERIENCE

Quantum Networks from Spin Model *Mar 2026 – present*

University of Pittsburgh · Advisors: Dr. Zhehao Dai & Dr. Kaushik Seshadreesan

- Investigating physical lattice models — specifically the Random Transverse-Field Ising Model (RTIM) — as a testbed for real-world quantum networks. Characterized the multipartite entanglement structure of the emergent quantum network, and currently developing methods to optimally exploit these entanglement resources by achieving network topologies suited to tasks such as quantum communication and routing, informed by the underlying spin model.

Hofstadter-Hubbard Model

May 2025 – Nov 2025

University of Pittsburgh · Advisor: Dr. Zhehao Dai

- Studied the fractional quantum Hall effect and Chern–Simons theory. Implemented a slave-rotor (parton) mean-field treatment of the triangular-lattice Hofstadter–Hubbard model, built a simulation of spinon-excitation dynamics, and probed the excitation's edge behavior on the lattice in Python.

Floquet Analysis for SNAIL

Aug 2024 – May 2025

University of Pittsburgh · Advisor: Dr. Roger Mong

- Derived the effective Floquet Hamiltonian of a driven anharmonic oscillator with cubic nonlinearity (the reduced model of a driven SNAIL circuit mode) via a Magnus expansion, and built a numerical time-evolution routine in Python to validate the derivation.

Master's Research Project — Diffusion under Stochastic Resetting

Jan 2022 – Jul 2022

IMSc Chennai · Advisor: Dr. Arnab Pal

- Studied escape properties of a diffusing particle in conical-tube geometry under a stochastic resetting strategy. Modified the Fick–Jacobs equation to include resetting and solved it analytically for the process's statistical quantities.
- Found escape times depend strongly on resetting rate and initial position; counter-intuitively, hindering the process at a finite rate can significantly speed up escape. Identified when this occurs and the optimal reset rate minimizing escape time. (*Published — see below.*)

INTERNSHIPS / PROJECTS

Reading Project — Adiabatic Gauge Potentials

Aug 2021 – Nov 2021

IIT Kanpur · Advisor: Dr. Arijit Kundu

- Studied the formalism of adiabatic gauge potentials (AGP) for classical/quantum systems, variational and perturbative methods to compute them in interacting systems, and their relation to the Fubini–Study metric, Berry curvature, and counter-diabatic driving. Explored AGPs as a probe of quantum phase transitions and quantum chaos, versus OTOCs and level statistics.

Summer Internship — Neutrino Phenomenology

Jun 2021 – Aug 2021

IISER Mohali · Advisor: Dr. Ambresh Shivaji

- Studied survival/transition probabilities of neutrino types vs. energy for different baseline approximations in 3+0 and 3+1 models, comparing against experimental data and reviewing discrepancies between current theories and experimental data motivating physics beyond the Standard Model.

PUBLICATIONS & PRESENTATIONS

- **S. Jain**, Z. Dai, K. P. Seshadreesan. “Multipartite entanglement characterization in quantum networks from spin clusters.” Poster accepted at IEEE QCE (Quantum Week) 2026; to appear in the Proceedings of IEEE QCE 2026. [\[extended abstract\]](#)
- P. Srivastava, A. R. Sabavat, **S. Jain**, A. Scheller-Wolf, S. Tayur, D. Tipper, P. Krishnamurthy, A. Babay, K. P. Seshadreesan. “Sequential vs. Simultaneous Entanglement Swapping under Optimal Link-Layer Control.” [arXiv:2605.04047](#)
- **S. Jain**, D. Boyer, A. Pal, L. Dagdug: “Fick–Jacobs description and first-passage dynamics for diffusion in a channel under stochastic resetting,” *Journal of Chemical Physics* (lead first author). doi.org/10.1063/5.0135249

TEACHING & SERVICE

- **Teaching Assistant / Fellow**: University of Pittsburgh — PHYS 0110/0111 (lab & theory), PHYS 0175 (lab & theory), PHYS 0525 (electronics lab), PHYS 0212/0219 (lab).
- **Referee**, Pittsburgh Interdisciplinary Mathematics Review (student-run peer-reviewed journal, Pitt & CMU).
- **Volunteer**, DPF-Pheno 2024 conference (Pitt & CMU).

TECHNICAL SKILLS

- **Programming**: Python, Mathematica, Julia, C++.
- **Computing**: Pitt Center for Research Computing (CRC / high-performance computing).

AWARDS/GRANTS & HONORS

- **PQI Community Collaboration Award** (2026) for *Dynamic Programming & Reinforcement Learning for Quantum Network Control (DyReQNet-Ctrl)* .
- **Academic Excellence Award**, IIT Kanpur (2020–21) — highest CGPA in the department.
- **National Topper** (top 25 of ~11,000) in 2020 and **State Topper** (top 1%) in 2019, National Graduate Physics Examination (NGPE) — a national undergraduate physics competition.
- **Madhava Mathematics Competition**: 1st state rank (2020), 3rd state rank (2019) — national undergraduate mathematics competition by HBCSE, Tata Institute of Fundamental Research.
- **Competitive entrance ranks**: All-India Rank 47/~15,000 (IIT-JAM), 46/~7,000 (JEST), and 161 (CSIR-NET JRF) — national exams for admission to Indian research institutes.
- UPSTSE Fellowship by the state government (2017).

SCHOOLS & WORKSHOPS

- **QNumerics Summer School**, College of Information & Computer Sciences, UMass Amherst (2026).
- **ICTS-TIFR summer courses**: “Light and Beyond” (Prof. R. Nityananda, 2020) and “Curves and Surfaces: Geometry and Physical Applications” (Prof. J. Samuel, 2022).
- Madhava Nurture Camp, St. Xavier's College Ahmedabad, National Board of Higher Mathematics(2019).